

Test 1: Balancing for 60+ Uninterrupted Seconds (seconds until fall)

Prototype	Trial 1 (s)	Trial 2 (s)	Trial 3 (s)	Trial 4 (s)	Trial 5 (s)	Result
Prototype 1	8	12	6	15	10	Fail
Prototype 2	20	18	25	22	19	Fail
Prototype 3	75	68	50	82	70	Success
Prototype 4	96	Indefinite (balanced)	102	Indefinite (balanced)	Indefinite (balanced)	Success

Note: Remaining trials ran for over 5 minutes and had no meaningful oscillations while maintaining balance: therefore, they can be considered "perfectly balanced"

Test 2: Recovers from Tap Interruptions

Prototype	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Result
Prototype 1	Fell Immediately	Overcorrected, Fell	Lag in Response, Fell	Overcorrected, Fell	Lag in Response, Fell	Fail
Prototype 2	Lag in Response, Fell	Overcorrected, Fell	Overcorrected, Fell	Lag in Response, Fell	Lag in Response, Fell	Fail
Prototype 3	Overcorrected, Fell	Overcorrected, Fell	Lag in Response, Fell	Lag in Response, Fell	Lag in Response, Fell	Fail
Prototype 4	Overcorrected, Fell	Recovered, brief wobble before stabilizing	Recovered, brief wobble before stabilizing	Recovered, brief wobble before stabilizing	Lag in Response, Fell	Partial Success

Test 3: Multi-Terrain Functionality

Prototype	Trial 1: Even Hardwood	Trial 2: Even Carpet	Trial 3: Uneven/Bumpy Carpet
Prototype 1	Fail	Fail	Fail
Prototype 2	Fail	Fail	Fail
Prototype 3	Pass	Fail	Fail
Prototype 4	Pass	Pass	Pass